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| **Fiscal Year:** | FY 2024 | **Task Last Updated:** | FY 06/07/2023 |
| **PI Name:** | Espinosa, Araceli Ph.D. | | |
| **Project Title:** | The Impact of Real Microgravity on the Proliferation of Human Neural Stem Cells and derived-Oligodendrocytes | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | SPACE BIOLOGY--Cellular and molecular biology | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology | | |
| **Space Biology Cross-Element Discipline:** | (1) Neurobiology | | |
| **Space Biology Special Category:** | (1) Cell Culture<br>(2) Translational (Countermeasure) Potential | | |
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| **Project Type:** | Flight | **Solicitation / Funding Source:** | 2014 Space Biology Flight NNH14ZTT001N |
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| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
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| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA ARC |
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| **Flight Program:** | ISS | | |
| **Flight Assignment:** | ISS<br>NOTE: End date changed to 11/30/2022 per F. Hernandez/ARC. (Ed. 6/20/2022)<br>NOTE: End date changed to 06/30/2022 per F. Hernandez/ARC and NSSC information. (Ed. 11/2/2021)<br>NOTE: Extended to 10/31/2021 per F. Hernandez/ARC and NSSC information (Ed., 11/19/2020)<br>NOTE: Extended to 10/31/2020 per F. Hernandez/ARC and NSSC information (Ed., 2/4/2020)<br>NOTE: Extended to 5/31/2020 per F. Hernandez/ARC and NSSC information (Ed., 6/12/19)<br>NOTE: Extended to 11/30/2019 per F. Hernandez/ARC (Ed., 10/24/18)<br>NOTE: Extended to 10/31/2018 per F. Hernandez/ARC (Ed., 11/2/17) | | |
| **Key Personnel Changes/Previous PI:** | August 2019 report: no changes have occurred. | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | NNX15AB43G | | |

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| **Performance Goal No.:** | |
| **Performance Goal Text:** | |
| **Task Description:** | Recent data from NASA shows that astronauts present vision, physiological, and anatomical changes as well as intracranial pressure (ICP) and visual impairment intracranial pressure (VIIP) (more recently known as Spaceflight Associated Neuro-ocular Syndrome (SANS)) after long-duration space missions. These changes are unlike any clinical entity previously seen. Yet, the cause(s) of chronically elevated ICP/VIIP and potential consequences are unknown. The main goal of this International Space Station (ISS) spaceflight proposal is to obtain reference data on the effects of real microgravity (0G) to validate and expand our recent data obtained using simulated microgravity (sim-μG). We want to assess the impact of real microgravity on the cell cycle of human neural stem cells and derived oligodendrocytes (OL) that are the myelin forming cells in the central nervous system. We now have evidence that sim-μG increases the number of neural progenitors and OL progenitor (OLP) via cell proliferation.<br>Here we postulate that microgravity-associated VIIP and altered ophthalmic anatomy (papilledema, choroidal folds, etc.) as well as a potential microgravity-associated ICP without visual impairment might be at least in part due to increased cell proliferation in the brain.<br><br>Aim I. To test the hypothesis that neural stem cells (NSC) and OL progenitors exposed to real microgravity will proliferate faster and will give rise to more progenies than cells grown in 1G.<br><br>Aim II. To test the hypothesis that in real microgravity NSC and OL produce and secrete one or more molecules that may act as regulators of cell proliferation.<br><br>Significance. Neural cells possess intrinsic plasticity that can be modulated by the environment and their response to real microgravity may confirm and even reveal new properties of these cells triggered by it. We also propose to examine molecules secreted by these cells while in space in order to identify potentially unique regulators of cell proliferation. It is known that secretome proteins play a key role in cell signaling and cell migration. These molecules may also hold a potential key for the modulation of normal cell proliferation for future therapeutic interventions for the central nervous system (CNS).<br><br>Approach. Human NSC and OLP will be plated using our proprietary specific culture media in self-contained hardware designed for 0G (Astrium). Control cells will be maintained in 1G.<br><br>Ground based experiments. We will analyze the effects of real microgravity on NSC and OL cell cycle via immunocytochemistry. After landing and recovery of the samples, the conditioned culture medium will be collected, maintained at 4°C, and transported to our laboratory to further characterize the secretome of both cell types maintained in real microgravity. Our stem cell culture medium as well as the oligodendrocyte specification medium are both serum-free and contain few proteins, properties that will facilitate the secretome analysis. The characterization of the secretome will include identification of proteins and small molecules that may be produced and secreted in real microgravity only.<br><br>Innovation. This is the first study on the impact of real microgravity on human neural cells. Our data will provide valuable information to better understand the gravitational epigenetic modulation of human neural cells.<br><br>Benefits for Humanity. Confirming that real microgravity increases cell proliferation will move us forward toward a better understanding on the causes of ICP and VIIP in astronauts. Moreover, since long-term missions in space are imminent, the identification of molecules that increase cell proliferation might lead to the design of potential countermeasures to prevent or alleviate the negative effects observed in astronauts exposed to long-term missions in space. Myelin is responsible for the rapid conduction of electrochemical messages to and from the brain. Since there is no cure for myelin disorders our data might also lead to novel therapeutic modalities for myelin repair. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | To date, there is no cure for developmental disabilities, neurodegenerative disorders, myelin loss, or cancer. We expect that our work described in the current proposal will generate cures that are of paramount importance in two different but complementary fields:<br>1) Drug discovery: The first product will be the first molecule that would have been identified, isolated, and tested to be responsible for increased cell proliferation. This will open the door for a second product (or series of products) under this category that would be an antibody or similar molecule(s) that would void/neutralize the effects of the first molecule on cell proliferation. This second product will be further studied as an anti-cancer agent via prevention of cell proliferation. If there is more than one relevant secreted molecule a similar process would be followed. Once we have data showing that we have interesting molecules, we will submit an invention report to the Office of Intellectual Property at UCLA (University of California, Los Angeles). This office supports scientists by contacting potentially interested companies and helping disseminate information on technologies available for licensing. In some cases UCLA technologies are commercialized by large companies. Alternatively, some entrepreneurs-inventors form successful startups based on their own UCLA technology.<br><br>2) Generation of cells for cell replacement therapies: The presence of functional myelin is essential for all our normal daily activities. Our overall goal is to investigate the prospects of cell replacement therapies for patients that require normal OL and myelin to restore proper CNS function. We contemplate to be able to apply this approach to propagate OLPs also known as (OPCs) and make them available to treat the spectrum of myelin disorders, from white matter damage in the premature neonate and infants suffering of autism or maternal prenatal infection, to the patient with leukodystrophies, multiple sclerosis, psychosis/schizophrenia, and trauma. We have previously demonstrated that OLPs are cells that, once grafted, migrate, integrate, and interact with the host CNS. Validating the functionality of OLPs exposed to real μG will open a new era on the ways we conceive cell replacement therapies and CNS repair. Moreover, trauma to the brain and spinal cord involves myelin loss and glial and neuronal cell death; thus, data generated by the current study will provide insights on producing cells to be transplanted into the CNS of individuals suffering from trauma.<br><br>Prior to commercializing the generation of cell populations in microgravity, it is essential to confirm that microgravity-exposed cells are fully functional and safe; by biochemical and pre-clinical neural transplantation proposed in this grant application we will generate the necessary evidence. Once this step will have been accomplished, we'll |

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| | contact specific companies that might provide us with the revenue necessary to fund the scale-up process to generate human induced pluripotent stem cell (hiPS) derived oligodendrocytes progenitors (OLP). By then we hope to have created strong partnership with the industry in order to be able to generate cells to overcome neural degeneration and developmental disabilities. |
| **Task Progress:** | Intracranial pressure (ICP) and intraocular pressure (IOP) are among the top health risks associated with long-term space travel. It has been proposed that in microgravity, fluids redistribute, increasing both ICP and OCP leading to visual impairment (VI) after spaceflight. Our original proposal aimed to test the hypothesis that space microgravity increases the proliferation of human neural stem cells (NSCs) and oligodendrocyte progenitors (OLPs), as we previously found using simulated microgravity.<br>This has been an inspirational journey; the Bioscience-4 research project consisted of two spaceflights. Using normal human brain cells, we explored cell division and maintenance of pluripotency for NSCs (the precursors of all cell types that form the brain and spinal cord). We examined the molecules they produced while in the Space Station and their behavior after spaceflight once we brought them back to Earth. Besides studying NSCs, we also ascertained the impact of spaceflight on OLPs, their survival in flight and post-flight, and the specific features that changed in the profile of molecules secreted by them while in space. We also examined their behavior post-flight.<br><br>Here we report great news: both cell types (NSCs and OLPs), survived the roundtrip and preserved their inherent properties. In the case of NSCs, they continued to “self-renew” after spaceflight, remaining pluripotent. This is important because NSCs give rise to neurons for learning and memory in the brain. This augurs well for a healthy brain in our astronauts, ensuring normal cognition and behavior. NSCs also generated astrocytes, one type of glial cell in charge of maintaining the function of the central nervous system (brain and spinal cord). The second cell type we sent to space, “oligodendrocytes”, which are the myelin-forming cells in the central nervous system (CNS), also survived and kept their properties in terms of proliferation (cell division) and expression of specific markers, confirming that they remained functional. We found that most cells sent to space remained normal, with the main difference of giving rise to more progenies than those kept in Earth’s gravity.<br><br>We also compared the molecules secreted by space (SPC)-flown NSCs and OLPs vs. those produced by their ground control counterparts (GC). We were surprised to see that, in general, changes were not as numerous as we expected; several pathways yielded very similar results between GC vs. space-flown cells, indicating that microgravity did not alter the cells’ natural properties. This is also reassuring for long-term space missions to the Moon and Mars. For other pathways, differences were small but significant regarding the abundance of proteins and small molecules between GC and space samples. Both the proteomics and metabolomics analyses allowed us to identify enhanced energetics activity in both NSCs and OLPs that flew to space. Such an increase in energy demands might be caused by cell proliferation. We are happy to report that the defense mechanisms of our cells were in place and responded adequately. For example, typical indicators of cellular stress are the increased expression of stress proteins, also known as heat shock proteins. Although they indicate cellular stress, these proteins are in charge of clearing misfolded proteins, allowing the cell to recover from the stress and synthesize normal proteins for their function. In other words, they indicate the activation of different mechanisms supporting survival and homeostasis in response to the stress produced by spaceflight.<br><br>We also found a decreased expression of proteins involved in visual function. Transthyretin (TTR) was reduced 4.6, and Calmodulin-2 decreased by 6.1 times in the SPC-NSCs secretome. The most significantly associated pathways include the metabolism of vitamins and cofactors related to visual phototransduction. These findings are important in that, by addressing these changes, visual functions can be maintained. Identifying these changes is just the first step towards mitigating or reversing them.<br><br>We also found a modest but significant increase in autophagy (paper in preparation), and a fraction of these cells presented larger size, all mechanisms reminiscent of the development of the CNS. Thus, it is not always abnormal, but somewhat necessary to find these behaviors in some cells, particularly when adapting to a different environment. It is important to note that, at present, we interpret these changes as “stress responses” because nobody has defined how to start identifying “non-stress” changes when referring to cellular responses specific to weightlessness or other gravitational changes. Considering the enormous difference between Earth’s gravity and microgravity, these differences may be suitable and normal. The changes observed by us and others studying life cells exposed to microgravity could also be transient and necessary to maintain both function and homeostasis in the brain. Within their context, they might be considered “normal and necessary” for a healthy transition from Earth’s gravity to different gravity – such as space, the Moon, or Mars.<br><br>As expected in all pioneer work, our study has some inherent limitations. Although we primarily focused on the effects of microgravity/spaceflight on NSCs and OLPs, other factors may contribute to the changes we observed. Therefore, future studies will have to examine the effects of radiation on both cell types’ proliferation and growth in our SPC-cells when compared to ground controls. For example, it is worth mentioning that experimental studies have shown that high atomic number energy (HZE) nuclei produce both qualitative and quantitative differences in biological effects compared to terrestrial radiation, leading to uncertainty in predicting exposure to health outcomes in humans. Despite all the unknowns, we feel that the key to the modulation of pathways and cell interactions may reside in the changes we observed. Nonetheless, the future for humankind in space appears bright as our work, and that of many other scientists, are identifying molecules and pathways as targets to design preventive measures and countermeasures for our current, future, and past astronauts. |
| **Bibliography Type:** | Description: (Last Updated: 02/07/2024) |
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